Section 7.2 Riemann Integrable Functions

We begin with a proof of the important Cauchy Criterion. We will then prove the Squeeze Theorem, which will be used to establish the Riemann integrability of several classes of functions (step functions, continuous functions, and monotone functions). Finally we will establish the Additivity Theorem.

We have already noted that direct use of the definition requires that we know the value of the integral. The Cauchy Criterion removes this need, but at the cost of considering two Riemann sums, instead of just one.

7.2.1 Cauchy Criterion *A function:* $[a, b] \to \mathbb{R}$ *belongs to* $\mathcal{R}[a,b]$ *if and only if for every* $\varepsilon > 0$ *there exists* $\eta_\varepsilon > 0$ *such that if* $\dot{\mathcal{P}}$ *and* $\dot{\mathcal{Q}}$ *are any tagged partitions of* $[a, b]$ *with* $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ *and* $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$, *then*

$$\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| < \varepsilon. \qquad \square$$

Proof. ($\Rightarrow$) If $f \in \mathcal{R}[a, b]$ with integral L, let $\eta_\varepsilon := \delta_{\varepsilon/2} > 0$ be such that if $\dot{\mathcal{P}}$, $\dot{\mathcal{Q}}$ are tagged partitions such that $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$, then

$$\left|S(f;\dot{\mathcal{P}}) - L\right| < \varepsilon/2 \quad \text{and} \quad \left|S(f;\dot{\mathcal{Q}}) - L\right| < \varepsilon/2.$$

Therefore we have

$$\begin{aligned}\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| &\le \left|S(f;\dot{\mathcal{P}}) - L + L - S(f;\dot{\mathcal{Q}})\right| \\ &\le \left|S(f;\dot{\mathcal{P}}) - L\right| + \left|L - S(f;\dot{\mathcal{Q}})\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

($\Leftarrow$) For each $n \in \mathbb{N}$, let $\delta_n > 0$ be such that if $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ are tagged partitions with norms $< \delta_n$, then

$$\left|S(f;\dot{\mathcal{P}}) - f(f;\dot{\mathcal{Q}})\right| < 1/n.$$

Evidently we may assume that $\delta_n \ge \delta_{n+1}$ for $n \in \mathbb{N}$; otherwise, we replace δ_n by $\delta_n' := \min\{\delta_1, \ldots, \delta_n\}$.

For each $n \in \mathbb{N}$, let $\dot{\mathcal{P}}_n$ be a tagged partition with $||\dot{\mathcal{P}}_n|| < \delta_n$. Clearly, if $m > n$ then both $\dot{\mathcal{P}}_m$ and $\dot{\mathcal{P}}_n$ have norms $< \delta_n$, so that

$$\left|S(f;\dot{\mathcal{P}}_n) - S(f;\dot{\mathcal{P}}_m)\right| < 1/n \qquad \text{for} \qquad m > n. \tag{1}$$

Consequently, the sequence $\left(S(f;\dot{\mathcal{P}}_m)\right)_{m=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$. Therefore (by Theorem 3.5.5) this sequence converges in $\mathbb{R}$ and we let $A := \lim_m S(f;\dot{\mathcal{P}}_m)$.

Passing to the limit in (1) as $m \to \infty$, we have

$$\left|S(f;\dot{\mathcal{P}}_n) - A\right| \le 1/n \qquad \text{for all} \qquad n \in \mathbb{N}.$$

To see that A is the Riemann integral of f, given $\varepsilon > 0$, let $K \in \mathbb{N}$ satisfy $K > 2/\varepsilon$. If $\dot{\mathcal{Q}}$ is any tagged partition with $||\dot{\mathcal{Q}}|| < \delta_K$, then

$$\begin{aligned}\left|S(f;\dot{\mathcal{Q}}) - A\right| &\le \left|S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{P}}_K)\right| + \left|S(f;\dot{\mathcal{P}}_K) - A\right| \\ &\le 1/K + 1/K < \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, then $f \in \mathcal{R}[a, b]$ with integral A. Q.E.D.

We will now give two examples of the use of the Cauchy Criterion.

7.2.2 Examples **(a)** Let $g : [0,3] \to \mathbb{R}$ be the function considered in Example 7.1.4(b). In that example we saw that if $\dot{\mathcal{P}}$ is a tagged partition of $[0, 3]$ with norm $||\dot{\mathcal{P}}|| < \delta$, then

$$8 - 5\delta \leq S(g;\dot{\mathcal{P}}) \leq 8 + 5\delta.$$

Hence if $\dot{\mathcal{Q}}$ is another tagged partition with $||\dot{\mathcal{Q}}|| < \delta$, then

$$8 - 5\delta \leq S(g;\dot{\mathcal{Q}}) \leq 8 + 5\delta.$$

If we subtract these two inequalities, we obtain

$$\left|S(g;\dot{\mathcal{P}}) - S(g;\dot{\mathcal{Q}})\right| \leq 10\delta.$$

In order to make this final term $< \varepsilon$, we are led to employ the Cauchy Criterion with $\eta_\varepsilon := \varepsilon/20$. (We leave the details to the reader.)

(b) The Cauchy Criterion can be used to show that a function $f : [a,b] \to \mathbb{R}$ is *not* Riemann integrable. To do this we need to show that: *There exists* $\varepsilon_0 > 0$ *such that for any* $\eta > 0$ *there exists tagged partitions* $\dot{\mathcal{P}}$ *and* $\dot{\mathcal{Q}}$ *with* $||\dot{\mathcal{P}}|| < \eta$ *and* $||\dot{\mathcal{Q}}|| < \eta$ *such that* $\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| \geq \varepsilon_0$.

We will apply these remarks to the Dirichlet function, considered in 5.1.6(g), defined by $f(x) := 1$ if $x \in [0,1]$ is rational and $f(x) := 0$ if $x \in [0,1]$ is irrational.

Here we take $\varepsilon_0 := \frac{1}{2}$. If $\dot{\mathcal{P}}$ is any partition all of whose tags are rational numbers then $S(f;\dot{\mathcal{P}}) = 1$, while if $\dot{\mathcal{Q}}$ is any tagged partition all of whose tags are irrational numbers then $S(f;\dot{\mathcal{Q}}) = 0$. Since we are able to take such tagged partitions with arbitrarily small norms, we conclude that the Dirichlet function is *not* Riemann integrable. □

The Squeeze Theorem

In working with the definition of Riemann integral, we have encountered two types of difficulties. First, for each partition, there are infinitely many choices of tags. And second, there are infinitely many partitions that have a norm less than a specified amount. We have experienced dealing with these difficulties in examples and proofs of theorems. We will now establish an important tool for proving integrability called the Squeeze Theorem that will provide some relief from those difficulties. It states that if a given function can be "squeezed" or bracketed between two functions that are known to be Riemann integrable with sufficient accuracy, then we may conclude that the given function is also Riemann integrable. The exact conditions are given in the statement of the theorem. (The idea of squeezing a function to establish integrability led the French mathematician Gaston Darboux to develop an approach to integration by means of upper and lower integrals, and this approach is presented in Section 7.4.)

7.2.3 Squeeze Theorem *Let* $f : [a,b] \to \mathbb{R}$. *Then* $f \in \mathcal{R}[a,b]$ *if and only if for every* $\varepsilon > 0$ *there exist functions* α_ε *and* ω_ε *in* $\mathcal{R}[a,b]$ *with*

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x) \qquad \text{for all} \qquad x \in [a,b], \tag{2}$$

and such that

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon. \tag{3}$$

Proof. ($\Rightarrow$) Take $\alpha_\varepsilon = \omega_\varepsilon = f$ for all $\varepsilon > 0$.

($\Leftarrow$) Let $\varepsilon > 0$. Since α_ε and ω_ε belong to $\mathcal{R}[a,b]$, there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$ then

$$\left|S(\alpha_\varepsilon;\dot{\mathcal{P}}) - \int_a^b \alpha_\varepsilon\right| < \varepsilon \qquad \text{and} \qquad \left|S(\omega_\varepsilon;\dot{\mathcal{P}}) - \int_a^b \omega_\varepsilon\right| < \varepsilon.$$

It follows from these inequalities that

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon;\dot{\mathcal{P}}) \qquad \text{and} \qquad S(\omega_\varepsilon;\dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

In view of inequality (2), we have $S(\alpha_\varepsilon;\dot{\mathcal{P}}) \leq S(f;\dot{\mathcal{P}}) \leq S(\omega_\varepsilon;\dot{\mathcal{P}})$, whence

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

If $\dot{\mathcal{Q}}$ is another tagged partition with $||\dot{\mathcal{Q}}|| < \delta_\varepsilon$, then we also have

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot{\mathcal{Q}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

If we subtract these two inequalities and use (3), we conclude that

$$\begin{aligned} |S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})| &< \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon \\ &= \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < 3\varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, the Cauchy Criterion implies that $f \in \mathcal{R}[a,b]$. Q.E.D.

Classes of Riemann Integrable Functions

The Squeeze Theorem is often used in connection with the class of step functions. It will be recalled from Definition 5.4.9 that a function $\varphi : [a,b] \to \mathbb{R}$ is a **step function** if it has only a finite number of distinct values, each value being assumed on one or more subintervals of $[a,b]$. For illustrations of step functions, see Figures 5.4.3 or 7.1.4.

7.2.4 Lemma If J *is a subinterval of* $[a,b]$ *having endpoints* $c < d$ *and if* $\varphi_J(x) := 1$ *for* $x \in J$ *and* $\varphi_J(x) := 0$ *elsewhere in* $[a,b]$, *then* $\varphi_J \in \mathcal{R}[a,b]$ *and* $\int_a^b \varphi_J = d - c$.

Proof. If $J = [c,d]$ with $c \leq d$, this is Exercise 7.1.13 and we can choose $\delta_\varepsilon := \varepsilon/4$.

There are three other subintervals J having the same endpoints c and d, namely, $[c,d)$, $(c,d]$, and (c,d). Since, by Theorem 7.1.3, we can change the value of a function at finitely many points without changing the integral, we have the same result for these other three subintervals. Therefore, we conclude that all four functions φ_J are integrable with integral equal to $d - c$. Q.E.D.

It is an important fact that any step function is Riemann integrable.

7.2.5 Theorem *If* $\varphi : [a,b] \to \mathbb{R}$ *is a step function, then* $\varphi \in \mathcal{R}[a,b]$.

Proof. Step functions of the type appearing in 7.2.4 are called "elementary step functions." In Exercise 5 it is shown that an arbitrary step function φ can be expressed

as a linear combination of such elementary step functions:

$$\varphi = \sum_{j=1}^{m} k_j \varphi_{J_j}, \tag{4}$$

where J_j has endpoints $c_j < d_j$. The lemma and Theorem 7.1.5(a, b) imply that $\varphi \in \mathcal{R}[a, b]$ and that

$$\int_a^b \varphi = \sum_{j=1}^{m} k_j (d_j - c_j). \tag{5}$$

Q.E.D.

We illustrate the use of step functions and the Squeeze Theorem in the next two examples. The first reconsiders a function that originally required a complicated calculation.

7.2.6 Examples **(a)** The function g in Example 7.1.4(b) is defined by $g(x) = 2$ for $0 \le x \le 1$ and $g(x) = 3$ for $1 < x \le 3$. We now see that g is a step function and therefore we calculate its integral to be $\int_0^3 g = 2 \cdot (1 - 0) + 3 \cdot (3 - 1) = 2 + 6 = 8$.

(b) Let $h(x) := x$ on [0,1] and let $P_n := (0, 1/n, 2/n, \ldots, (n-1)/n, n/n = 1)$. We define the step functions α_n and ω_n on the disjoint subintervals $[0, 1/n)$, $[1/n, 2/n)$, $\ldots, [(n-2)/n), (n-1)/n)$, $[(n-1)/n, 1]$ as follows: $\alpha_n(x) := h((k-1)/n) = (k-1)/n$ for x in $[(k-1)/n, k/n)$ for $k = 1, 2, \ldots, n-1$, and $\alpha_n(x) := h((n-1)/n) = (n-1)/n$ for x in $[(n-1)/n, 1]$. That is, α_n has the minimum value of h on each subinterval. Similarly, we define ω_n to be the maximum value of h on each subinterval, that is, $\omega_n(x) := k/n$ for x in $[(k-1)/n, k/n)$ for $k = 1, 2, \ldots, n-1$, and $\omega_n(x) := 1$ for x in $[(n-1)/n, 1]$. (The reader should draw a sketch for the case $n = 4$.)

Then we get

$$\begin{aligned} \int_0^1 \alpha_n &= \frac{1}{n}(0 + 1/n + 2/n + \cdots + (n-1)/n) \\ &= \frac{1}{n^2}(1 + 2 + \cdots + (n-1)) \\ &= \frac{1}{n^2} \cdot \frac{(n-1)n}{2} = \frac{1}{2}(1 - 1/n). \end{aligned}$$

In a similar manner, we also get $\int_0^1 \omega_n = \frac{1}{2}(1 + 1/n)$. Thus we have

$$\alpha_n(x) \le h(x) \le \omega_n(x)$$

for $x \in [0, 1]$ and

$$\int_0^1 (\omega_n - \alpha_n) = \frac{1}{n}.$$

Since for a given $\varepsilon > 0$, we can choose n so that $\frac{1}{n} < \varepsilon$, it follows from the Squeeze Theorem that h is integrable. We also see that the value of the integral of h lies between the integrals of α_n and ω_n for all n and therefore has value $\frac{1}{2}$. □

We will now use the Squeeze Theorem to show that an arbitrary continuous function is Riemann integrable.